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(Key and Solutions for AIMCAT2408)

Key

SECTION – I

1. A
2. A
3. B
4. D
5. D
6. B
7. B
8. A
9. B
10. C
11. D
12. C
13. B
14. A
15. D
16. B
17. B
18. 3142
19. 1432
20. B
21. D
22. B
23. 2431
24. B

SECTION – II

1. A
2. C
3. B
4. D
5. A
6. B
7. C
8. A
9. D
10. D
11. 40
12. 20
13. 90
14. 85
15. D
16. A
17. 124
18. 18
19. A
20. D

SECTION – III

1. 73
2. B
3. C
4. 60
5. A
6. 48
7. 3
8. B
9. 72
10. B
11. A
12. C
13. 44
14. 8
15. 42
16. C
17. D
18. C
19. B
20. 144
21. 20.5
22. D

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 574

1. Option A: Sine qua non means an essential condition that is absolutely necessary. So, the author is clearly stating that concentration is absolutely mandatory for the game of chess and it is impossible to indulge in the experience, otherwise. Option A is the answer.
Option B: Concentration is required right through the game, and not just to perceive the beauty of good moves. Also, by talking about the beauty of good moves, the author is discussing the game but this is not the only thing he discusses in justifying that concentration is the sine qua non of the chess experience. Option B is not the answer.
Option C: While this is true, it is not the reason why the author calls concentration the sine qua non, as it is important to concentrate right through (and not just at the point of tactical contact). Option C is not the answer.
Option D: Once again, while this is true, winning and losing is not the only reason why the author thinks concentration is important. Concentration is the backbone of the whole exercise from start to finish. Option D is not the answer.

Choice (A)

2. Option A: The author likens thermostats to humans (using 'like') which makes it a simile. More importantly, the author calls humans thermostats to explain how we vary – receive and send signals – based on the ambience, while sometimes, but not always, adjusted against our will. So, thermostats are a simile for the way humans transact with our ambiences – every which way. Option A is the answer.
Option B: The example wasn't to articulate how humans are 'influenced' by whatever is surrounding us. We are affected, sure, but we also self-regulate. Option B is not the answer.
Option C: One object compared to another is not an analogy. One situation likened to another is an analogy. Also, the example of thermostats wasn't given to just discuss humans optimising outside situations. We do seek the 'optimal temperature' which is a metaphor for internal

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stability/concentration. However, we are not 'optimising' everything that is thrown at us. Option C is not the answer.
Option D: Humans have been called thermostats. Thermostats haven't been likened to our 'ability to regulate'. We have been called thermostats because we adjust internally to whatever is going outside, sometimes against our will. Hence, Option D is not the answer. Choice (A)

3. Option A: There are two problems. It is not only other phenomena which provide meta-cognitive views. Concentration can do that as well. Secondly, the author is saying concentration may be similar to other such phenomena. So, the inference that concentration is different is not valid. Option A is not the answer.
Option B: This is apt, because the author says that meta-cognitive views allow the mind to become aware of itself, which can be understood as a way of our mind understanding itself, or us understanding our mind/thoughts. Option B is the answer.
Option C: Meta-cognitive views are not activities. They are contexts of meaning and activity, which provide awareness of our mind. Hence, Option C is easy to eliminate.
Option D: The culmination of various aspects of our psyche results in concentration as understood from 'concentrating is more like a method of corralling and coordinating fissiparous parts of our psyche'. The author isn't talking about meta-cognitive views in the line being referenced. Option D is not the answer. Choice (B)

4. Option A: While flow involves the loss of self-consciousness, and flow isn't exactly a good model for being a lodestar for living, the author doesn't have a bone to pick against loss of self-consciousness. So, we cannot quite infer what his take is on the loss of self-consciousness and whether it is desirable or detestable. Option A is not the answer.
Option B: This represents a false dichotomy. While the author does suggest that virtue is important and that is what would be the lodestar for living, something that nurtures virtue, the author doesn't really have an issue with pleasure. Something that nurtures virtue for example, while offering pleasure wouldn't really be disapproved by the author. Option B is not the answer.
Option C: The author gives importance to both – commitment to meaning and appreciation of method. So, the lodestar for living wouldn't pit those two important aspects against each other, or push us to pick one over the other. Option C is not the answer.
Option D: The author opines that as a lodestar (guiding principle) for living flow has limitations, and goes on to explain that it doesn't help build desirable qualities in one's character. So, we could infer that the author's lodestar for living would be something that helps build good character. Hence, Option D is the answer. Choice (D)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 527

5. Option A: The last sentence of the passage indicates that the trilemma isn't always true. Helene Ray proved that unlike what the trilemma suggests, one change of the three parameters leaves both the others affected. Hence, Option A is not the answer.
Option B: While this is true, this is not the implication of the last line, which states that it is not just the floating currencies which adjust, but also the monetary policy – which isn't left undisturbed as the trilemma suggests it should have been. Option B is not the answer.
Option C: Domestic monetary policy does not prevent floating currencies from adjusting to capital flows. It is the other way around according to the last line of the passage. That the adjustment of the floating currency to capital flow doesn't leave the monetary policy untouched or unaffected. Option C is not the answer.
Option D: Capital flows lead to adjustment of floating currencies, but the floating currencies do not adjust in a way as to leave the monetary policy unsullied. In other words, the monetary policy is indeed affected as well (unlike what the trilemma suggests). Hence, Option D is the answer.
Choice (D)

6. Option A: S2 doesn't strengthen S1. It contradicts it. S1 states that when there are capital flows, a country can choose between A: monetary policy control and floating exchange rate, or B: fixed exchange rate and monetary bondage. S2 on the other hand states that even if a country chooses floating exchange rate, it cannot enjoy monetary policy control. Hence, Option A is not the answer.
Option B: If S2 is true, a country doesn't really have monetary policy control, when open to capital flows, even when it accepts a floating exchange rate. This straightaway contradicts what the trilemma suggests that in the presence of capital flows, floating exchange rate leads to monetary policy control. So, S2 invalidates S1. In other words, both the statements can't be true at the same time. Option B is the answer.
Option C: S2 is an argument rather than evidence (which is verifiable and hence, not an opinion or an inference). Also, S2 doesn't support S1. So, Option C is not the answer.
Option D: S2 contradicts S1, as explained in Option B. So, S2 cannot be arrived at using S1. Hence, Option D is not the answer. Choice (B)

7. Option A: The para doesn't give us any indication that the Netherlands benefited from the currency pegging. Monetary serfdom didn't affect the Netherlands, because of similar business cycles to Germany's. That's all we understand from the para. Hence, Option A is not the answer.

Option B: The example shows that Spain and Greece couldn't get themselves out of trouble because of the monetary serfdom that they had signed up for by pegging their currencies to that of Germany. That loss of monetary independence is what happens because of currency pegging. That is what has been suggested by the example. Hence, it is true that currency pegging doesn't leave elbow room or manoeuvring space for a country's monetary policy. Option B is the answer.
Option C: From the para it can be understood that we are talking about a phase when the currencies were pegged to the German currency. We cannot be sure that the crisis and problems encountered by Spain and Greece were after the shift to Euro (especially because the para starts with 'In the run-up). We are only discussing the consequences of pegging the currencies and losing monetary independence. Option C is not the answer.
Option D: Spain and Greece were mentioned as an example of how monetary serfdom could backfire. That doesn't really indicate that the Netherlands faced and then survived the same problems as these two countries. In other words, we cannot infer that the Netherlands was also hit by the same crisis. Hence, Option D is not the answer. Choice (B)

8. Option A: According to the policy trilemma argument, only two of the three parameters can be regulated. So, it is true that if a country doesn't regulate the exchange rate (allows it to float), it can enjoy capital mobility and monetary policy autonomy. So, Option A does reflect the logic of the Policy Trilemma. Option A is the answer.
Option B: If the exchange rate is fixed, and capital flow is allowed, then a country loses its monetary policy independence. So, this option doesn't reflect the Policy Trilemma. Option B is not the answer.
Option C: If there is capital mobility, a country cannot fix its exchange rate if it still wants to enjoy monetary autonomy. So, a country that has capital mobility (doesn't regulate it) shouldn't fix its exchange rate if it wants to control its monetary policy. The option contradicts the Policy Trilemma. Option C is not the answer.
Option D: Monetary autonomy is one of the three parameters. It isn't true that if this parameter is achieved the other two can be controlled as well. Only two of the three can be controlled. So, Option D is not the answer. Choice (A)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 549

9. Option A: The Third Way doesn't alternate between capitalist and communist ways. If something alternates between A and B, it means, it is sometimes A and sometimes B. The Third Way was neither capitalism nor communism. It was an ***alternative viewpoint***. Option A is not the answer.
Option B: Two statements from the passage give us information about The Third Way. '*Paris was the place where the world's most original voices of the time tried to find an independent and* ***original alternative to the capitalist and Communist models for life, arts, and politics*** *- a 'Third Way'.*' Also, '*they did all they could to question both US policies and the Communist Party's views. Paris was, for them, both a refuge and a bridge to think in a different way. They opened up the possibility of a Third Way.*' From these two statements it can be understood that the Third Way was an alternative perspective, that neither subscribed to communist philosophy nor followed the US capitalistic views. Option B is the answer.
Option C: While the philosophy of the Third Way was indeed influenced by the perils of World War II and the Nazi occupation, it wasn't a social model, per se. It was also political, and it was more a way of thinking than a working model. That makes this option inaccurate.
Option D: The distinction drawn in the passage was between capitalist and communist views. Here, it suggests that the distinction was between the policies of European Union and the US policies, which is not true at all. Also, the EU wasn't founded during the time frame when The Third Way was the idea being propagated, as understood from '*in what would later become the European Union*'. Option D is not the answer. Choice (B)

10. Option A: From the structure of the last line of the passage, this option can be eliminated. Because the sentence clearly says, it was as much about 'post-war Parisian intellectual irresponsibility' as it was about political, artistic, moral and sexual incandescence. So, these are two opposing ends. Hence, the incandescence is not what the intellectual irresponsibility refers to. Option A is not the answer.
Option B: This is a positive concept and hence, can be straightaway ruled out since the question asks us about intellectual irresponsibility and not about 'renaissance' that was brought about by the intellectuals of Paris. Hence, Option B is not the answer.
Option C: Consider the sentences: '*Tony Judt addresses the latter in his academic work Past Imperfect: French Intellectuals 1944–1956… Paris intellectuals had so much power bestowed on them by circumstances and their own genius and yet failed, in his view, to change the world.*' We can understand why Tony Judt called it intellectual irresponsibility from these statements. It was because Parisian intellectuals had a lot of power bestowed upon them (power in the form of intellectual authority and trust in their philosophies). Also, there was no doubt that these individuals were intellectually gifted. Yet, they didn't bring about the kind of change in the world as one would have hoped for or expected from them. In other words, it was a wasted opportunity.

That is what the author calls post-war Parisian intellectual irresponsibility. Option C is the answer.
Option D: Once again, this is a positive change brought about by the Parisian intellectuals. They ushered into an era of debates and progressive thinking along with politics that didn't subscribe to the norms. Philosophers indulging in politics and creating new ways of thinking has been given as an example in the passage of positive changes during the era. This cannot be dubbed as intellectual irresponsibility. Option D is not the answer. Choice (C)

11. Option A: The tone the author adopts here is one that patronises the Parisian intellectuals, defends them. The author seems to counter the allegation that the Parisian intellectuals were irresponsible. Instead, the author argues that they did something spectacular to even raise such hopes to be called irresponsible. This option suggests the reverse, that it counters the false sense of greatness. So, while it is almost a rhetorical question, the result is not the same as what the option supposes. Hence, Option A is not the answer.
Option B: While this option gets the positivity about the Parisian intellectuals right, there is nothing in the author's tone that suggests hyperbole and eulogising (excessive and often unreasonable praise). The author also doesn't 'discredit all criticism' against the intellectuals. All the author is trying to do is tone down the criticism, while not disagreeing with it entirely. Option B is not the answer.
Option C: There is no hint of sarcasm here as the author asks a rhetorical question intending something positive about the Parisian intellectuals. Option C is not the answer.
Option D: This is apt as the author does want to give credit to the Parisian intellectuals – credit where it is due. They deserve credit for the fact that they ignited hope, they raised the bar for expectations. Only because they raised the bar of expectations did we come to the point where someone felt they were intellectually irresponsible and didn't fulfil the promise. Hence, Option D is the answer. Choice (D)

12. Option A: This can be understood from '*but they had three things in common:* ***the experience of war****, their brush with death and the elation of the Liberation of Paris*' and '*After the horrors of war*
Option C: This is not true. It is not that the entire generation were migrants from elsewhere. Yes, there were the odd artist or two who were censored elsewhere and were adopted by Paris (Miller, for example, or black Jazz musicians). But, not all of them reached Paris. Some of them were from Paris. And some of them learnt to question the status quo in Paris. It is not true that they questioned norms elsewhere and then were chased out. Paris was a metaphorical refuge for thinkers, not literally a refuge for everyone. Hence, Option C is the answer.
Option D: This is true as all kinds of art, music and political philosophy thrived during that era, with open debates. This has been summarised in '*Paris was, for them, both a refuge and a bridge to think in a different way.*' Hence, Option D is not the answer. Choice (C)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 556

13. In the passage, the author defines the term "contagion" as "the risk of a country being affected by developments totally outside of its policy ambit".
Option A: In this option, China deciding to print new currency for circulation is affecting the U.S dollar rate. However, this is outside the control of U.S.'s domestic policy and so, it is a valid example of a contagion.
Option B: Since France is a part of the UN security council, it is possible that they may influence the decision of how of shipment of arms can flow in and out of France. So this is not exactly a case of France being affected by developments outside its policy gambit. This is not an example of a "contagion" and so, option B is the answer.
Option C: Pakistan's decision to build a chemical plant on their land and discharge waste into the Indus river is not something India's domestic policy can influence in any way and is a clear example of a contagion. C is not the answer.
Option D: Scientific expedition in Antarctica causing a spike in temperatures in the South Pole, impacting Australian ecosystem is an example of contagion since Australia's domestic policy cannot directly address this. Option D is also not the answer. Choice (B)

that shaped and informed them…'. So, we can say they were influenced by the trials and tribulations of war. Hence, Option A is not the answer.
Option B: This can be understood from '*Paris was the place where the world's most original voices of the time tried to find an independent and original alternative to the capitalist and Communist models for life, arts, and politics - a 'Third Way'.*' The Parisian intellectuals discussed in the passage were original, and open-minded as well, questioning the norms and status quo and developing their own independent worldview. Option B is not the answer.

14. An apt summary of the concept of globalisation with respect to public policy would incorporate the critical issue which is "*the trade-off between individual freedom and collective self-interest*", along with the definition of globalisation which is "*the movements of goods, people and services*".
Option A: This option defines the term 'globalisation' and as for the definition of globalisation with respect to public policy, it mentions that public policy curbs individual freedom. This is mentioned in the passage, and can be understood from the line "A nation-state is presumed to restrict its citizens' involvement…with only their collective self-interest in mind". The author mentions that while this could happen, it is important to strike a balance between the two. Hence, Option A is correct.
Option B: This option is partially correct – it's definition of globalisation is correct. There is no mention of public policy and also, it talks about curbing individual freedom, but does not mention the cause for this restriction, which is, collective self-interest. Hence, Option B is incorrect.
Option C: This option, on the surface, looks like the perfect summary. However, on delving deeper, we see that it mentions that globalisation causes the critical issue of the trade-off between collective self-interest and individual freedom. The passage only states that there is a critical issue in the context of public policy relating to globalisation. It does not mean that globalisation is what causes the critical issue. This is not correct and so Option C is not the answer.
Option D: Here, definition of globalisation is incorrect, it mentions that globalisation allows for intermingling of goods and services. However, the intermingling of said goods and services itself is globalisation. Hence, Option D is incorrect.
Choice (A)

15. On reading the passage thoroughly, we notice that there are certain points that the author has touched upon but has not discussed in detail. Studies related to such points will take forward the author's ideas in the passage.
Option A: The author mentions that "*a critical issue*" for globalisation "is the trade-off between individual freedom and collective self-interest, and also whether the burden of proof lies with the individual or with the national authorities". However, he does not take forward this discussion

Option D: Economic integration through the movement of goods, people and capital (globalisation) is something that the author has discussed in detail in the third para of the passage. Further studies about this concept will not add much value, since the author has already shed light on this concept in the passage. Hence, option D is the correct answer. Choice (D)

16. The author mentions that there are "wider macroeconomics implications" with regard to cross-border capital flows. The macroeconomic elements other than globalisation, to be impacted by exchange rates, are the monetary policy and financial system of a nation-state. The author has laid the groundwork for mentioning this impact on other macroeconomic elements by stating that, not just globalisation, but other elements as well are impacted by cross-border flows of capital.
Option A: The phenomenon of contagion "implies the risk of a country being affected by developments totally outside of its policy ambit". This is something that the author has already discussed in detail. Also, the author goes on to talk about what the macroeconomic implications are, and we see that they are in no way connected to the 'contagion'. Hence, Option A is incorrect.
Option B: This option talks about "the conduct of monetary policy and the stability of the financial system" which is exactly what the author wants to discuss when he mentions "wider macroeconomic implications". Hence, Option B is the correct answer.
Option C: The author mentions exchange rates to illustrate that cross-border flow of capital impacts globalisation and other elements, but this is an example and is not the actual point that the author is trying to make. Hence, Option C is not the answer.

in the passage. Resolving these issues will improve the process of globalisation. Hence, Option A is not the answer since it adds value to the discussion in the passage.

Option B: Similar to Option A, this option is somewhat touched upon in the second para of the passage and again towards the end. So, a study of political upheavals that impact capital flows will add value to the discussion in the passage (although the author had mentioned that he is leaving out political aspects of globalization, the effect of politics on capital flows will be relevant for the discussion in the passage). So, Option B is incorrect.

Option C: In the last para of the passage, the author introduces the concept of how cross-border capital flows have "wider macroeconomic implications" in terms of how exchange rates determine not just the costs of globalisation, but also "the conduct of monetary policy and the efficiency as well as stability of the financial system". Further analysis of this topic will add to the author's argument of cross-border capital flows having wider macroeconomic implications and so Option C is not the answer.

answer.

Option D: To "exemplify some of the macroeconomic elements outside the policy ambit of nation-states" means to provide examples for the phenomenon of contagion. This is not what the author is doing in the last para. Hence, Option D cannot be the answer. Choice (B)

Solution for question 17:

17. Option A: This doesn't represent the essence of the para as it leaves out information about what those assumptions are. Based on the first part of the para, we can understand that those assumptions are about what an individual's goal is. If that is known, then it can be understood whether the individual is rational, based on how he or she acts based on the knowledge he or she has about the situation. Option A is not the answer.

Option B: This represents the essence of the para as it conveys the main ideas: that public choice approach considers rationality subjective (dynamically dependent), and dependent on whether an individual's actions are aligned to his or her goals based on their information about their situation. That correlation is what defines a person's rationality according to this approach.

Hence, Option B is the answer.

Option C: Public choice theorists do not make rational assumptions about an individual's situation. The assumptions are made about an individual's aims and the individual's knowledge of their situation. This option completely misrepresents the public choice approach. Option C is not the answer.

Option D: This changes the meaning of the para by introducing the term 'goal-oriented'. The para doesn't state that only a goal-oriented person can be rational. It might be quite likely that not pursuing a certain goal given the knowledge of the situation could be a more rational thing to do. Hence, Option D is not the answer. Choice (B)

Solutions for questions 18 and 19:

18. Sentence 1 is a downstream argument (indicated by 'then') that points out that forerunners in civilisation did not possess the idea of history.

Sentence 2 is a dependent statement, as it gives the cause using 'because' pointing to a previous sentence. It also refers to plural nouns using 'they'.

Sentence 3 is an independent sentence that talks about the historical consciousness of the ancient Sumerians.

Sentence 4 points to a previous sentence using 'this' but the bigger clue is what follows. It says, 'This was not because of x'. Such a sentence is obviously followed by something like 'This was because of y'. So, we know that 42 is a mandatory pair.

elemental maps. Also, 2 has a sense of conclusion when it uses 'showing that' to explain how authigenic mineralisation helped in preservation of the fossils. So, 1432. Ans: (1432)

Solutions for questions 20 and 21:

20. The statement immediately following blank (1) talks about alternatives. These alternatives expand on "alternative forms" mentioned in the statement immediately preceding blank (1). Therefore, these two sentences are connected and inserting the highlighted sentence between them would break the flow. Hence, the highlighted sentence does not fit into blank (1).

"The unsurprising result" in the statement immediately following blank (2) has to be the result of what is mentioned in the highlighted sentence – intensified preference for hierarchy and concentrated leadership. The statement immediately preceding blank (2) does not talk about any action and therefore, cannot result in anything. Therefore, there has to be a bridge between this statement and the statement immediately following blank (2). Hence, the highlighted sentence fits into blank (2).

"These choices" in the statement immediately following blank (3) refers to the "increased economic coordination in large firms and disfavoured economic cooperation between small players. Therefore, these two statements are connected. Additionally, the highlighted sentence

pair.
Now sentence 3 itself doesn't tell us what is an occult entity or what is Occam's Razor. But, we can definitely conclude that 1 comes after 3. That's because 42 are both explaining why this is the case (this is not because of this, but because of that). What 42 refer to as 'this' and 'it' is the absence of the idea of history. So, 3142 is the order. Ans: (3142)

19. Sentence 1 is an independent sentence. Don't confuse 'although' with a connector; it isn't. It only points to the contrast within the sentence, not outside it. It introduces the disagreement on the mode of preservation of the fossils after burial.
Sentence 2 connects to a preceding sentence through 'these fossils' explaining how the fossils were preserved. So, we can understand that 1 is upstream to 2.
Sentence 3 refers to 'differences' mentioned in the preceding sentence, explaining the contrast in the composition of the minerals.
Sentence 4 is an independent sentence that connects elemental mapping and differences in anatomical features.
We can understand that 43 is a block because 4 introduces variations (of relative abundance of elements) while 3 points to 'these differences'. So, we know that 143 is the general skeleton. The big question is where to put 2. 2 cannot precede the 43 block because 2 is downstream to 4. It is 4 that introduces the concept of elemental mapping, so it cannot be preceded by 2 which talks about

mentions only one preference while the statement immediately following blank (3) mentions more than one preference (choices). Hence, the highlighted sentence does not fit into blank (3).
The statement immediately following blank (4) mentions "the choice" and the statement immediately preceding blank (4) states what the choices are. Therefore, these two sentences are connected and inserting the highlighted sentence, which talks about antitrust laws and institutions preferring hierarchy, between them would break the thought flow. Hence, the highlighted sentence does not fit into blank (4). Choice (B)

21. If we focus only on the downside mentioned in the highlighted sentence, all the blanks seem like the right answer. An easier way to answer this would be to focus on "worst of all".
The highlighted sentence mentions one of the downsides of "it." It also states that it is the worst of all. Hence, it should be preceded by a statement that mentions at least one downside of "it." The statement immediately preceding blank (1) states that the uncritical worship of the language used to model stars has a downside. It does not explain what the downside is. Hence, the highlighted sentence does not fit into blank (1).
The statement immediately preceding blank (2) mentions Paul Romer bringing to attention the issue of mathiness. This too does not include any downside. Hence, the highlighted sentence does not fit into blank (2).

The statement immediately preceding blank (3) also does not mention any downside. It only states that macroeconomics is plagued by mathiness but does not explain what the negative side of it is. Hence, the highlighted sentence does not fit into blank (3).
The statement immediately preceding blank (4) mentions some downsides of the ubiquity of mathematical theory in economics and the highlighted sentence, which states the worst downside, can follow it. Also, it can be understood that "it" in the highlighted sentence refers to the ubiquity of mathematical theory in economics. Hence, the highlighted sentence fits into blank (4). Choice (D)

Solution for question 22:

Sentence 3 starts with a contrast marker 'But' mentioning the rise and fall of Wari and introducing a puzzle–the overlap of the collapse of the state with a drought.
Sentence 4 references a previous sentence where some sort of innovative engineering has been discussed.
It is clear that 4 is pointing to 2 where innovative engineering has been described (Canals and aqueducts) 'such' helping us put 4 after 2. So, 24 is a block. 1 has to follow 3 as it elaborates on what the puzzle is. Besides, 3 starts with 'But' so it has to come before 1, which has the same tone. 31 is a block, different in tone from the positive one in 24. So, 2431 is the order. Ans: (2431)

Solution for question 24:

22. Option A: While a part of this option is indeed true, the option, notwithstanding its rhetorical text, also focuses on the job or duty of a historian. The essence of the para is more about the notion of an impartial global history, that deviates from an earlier Eurocentric view of political and economic history, which as subjects themselves were quite partial towards one region. Option A is not the answer.

Option B: Eurocentric historians were wrong about favouring one region over another. The framework of global history avoids these wrongs, predominantly the undermining of the developments in non-European regions. The notion of global history moves away from the Eurocentric idea of modern development, to ensure that one region isn't privileged over another. Option B is the answer.

Option C: This option is true, except it doesn't reflect the entire essence of the para. The main idea in the para is that of the notion of 'global history' which doesn't favor one region over the other, something historians have been guilty of. Option C is not the answer.

Option D: This can be ruled out immediately. The para doesn't really mention that historiography is trying to correct the Eurocentric view. The para starts with a question – the status of 'global history'. Secondly, focusing on global history hasn't been mentioned in the para as a way of getting rid of the Eurocentric view (that nuance is important). All the para is saying is, that in the past historians were guilty of a Eurocentric view and global history tries to not favor one region over the other. Option D is not the answer.

Choice (B)

Solution for question 23:

23. Sentence 1 is a grammatically independent sentence, but contextually dependent (what society are we talking about?)

Sentence 2 is an independent sentence that explains how colonists from Wari created a network of canals in Moquegua Valley.

24. Option A: The para doesn't pit consciousness versus economics directly. It also doesn't really negate the influence of consciousness on social political and legal structures (a misrepresentation of the sub-ideas in itself). Consciousness wasn't discussed really in that capacity. It was only compared with social existence and its causal relationship with social existence was discussed. Option A can be eliminated.

Option B: This mentions both the main ideas of the para: that the material forces of production constitute the economic structure of society, and that the social, political (along with legal) and intellectual life (consciousness) is dependent on this economic structure. This represents the essence of the para. Hence, Option B is the answer.

Option C: While the para mentions that the political and legal superstructures are based on the economic structure, it doesn't really connect the political and legal superstructures to will and doesn't even mention that 'they are not based on their will.' The author talks about will only to show how it doesn't influence the relations of material production. Option C is not the answer.

Option D: It is not economics but 'the economic structure of society' which is the accumulation/sum total of the relations of production. Hence, Option D can be eliminated.

Choice (B)

Difficulty level wise summary - Section I

Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	5, 7, 10, 12, 21
Difficult	2, 6, 8, 9, 11, 13, 14, 15, 16, 17, 18, 20, 23
Very Difficult	1, 3, 4, 19, 22, 24

SECTION – II

Solutions for questions 1 to 5:

In 2020, one of the companies had an expenditure of USD 80 mn. Hence, in 2021, the expenditure of this company would have been 80 × 1.2 = USD 96 mn and in 2022, the expenditure of this company would have been USD 115.2 mn.
In 2021, one of the companies had an expenditure of USD 102 mn. In 2020, the expenditure of this company would have been 102/1.2 = USD 85 mn and in 2022, the expenditure of this company would have been USD 122.4 mn.
In 2022, one of the companies had an expenditure of USD 108 mn. In 2021, the expenditure of this company would have been USD 90 mn and in 2020, the expenditure of this company would have been USD 75 mn.
From (vi), we can find the expenditures of the other two companies in the three years (the two companies mentioned in (vi) are not the same as the ones mentioned in (iii), (iv) and (v), since their expenditures are different).
The following table provides the revenue of the three companies in the three years:

	2020	2021	2022
Company 1	80	96	115.2
Company 2	85	102	122.4
Company 3	75	90	108
Company 4	81.67	98	117.6
Company 5	83.33	100	120

In 2020, the company with USD 80 mn expenditure had the highest profit percentage.
The maximum possible profit percentage of this company is $\frac{150-80}{80} = 0.875$
Hence, the company with expenditure of USD 85 mn must have a revenue less than 85 × (1 + 0.875) = 159.375 (as the profit percentage should not exceed 87.5%).
Similarly, the company with expenditure of USD 75 mn must have a revenue less than 75 × (1 + 0.875) = 140.625.
Hence, Company 3 must be R or S or T.
In 2021, Company 2, which has USD 102 mn expenditure, had the highest profit percentage.
The maximum possible profit percentage of this company can be $\frac{220-102}{102} = 1.157\ or\ 115.7\%$

Company 1, which has expenditure of USD 96 mn, must have revenue less than 96 × 2.157 = 207.072. Hence, this company can only be Q or T.
Company 3, which has expenditure of USD 90 mn, must have revenue less than 90 × 2.157 = 194.13. Hence, this company also can only be Q or T.
From above, we have that Company 3 can only be R or S or T. Hence, **Company 3 must be T**.
Since Company 3 is T, **Company 1 must be Q**.
Also, for Company 4, the revenue must be less than 98 × 2.157 = 211.386. Hence, Company 4 can only be P or Q or T. Since Company 3 is T and Company 1 is Q, **Company 4 must be P**.
The revenue of Company 5 must be less than 100 × 2.157 = 215.7. Hence, **Company 5 must be S**.
The remaining company, **Company 2 must be R**.
The following table provides the revenue and expenditure of the five companies in each of the three years:

Company	2020		2021		2022	
	Revenue	Expenditure	Revenue	Expenditure	Revenue	Expenditure
P	145.0	81.67	210.0	98.0	210.0	117.6
Q	150.0	80.0	180.0	96.0	190.0	115.2
R	140.0	85.0	220.0	102.0	180.0	122.4
S	110.0	83.33	215.0	100.0	150.0	120
T	120.0	75.0	190.0	90.0	220.0	108.0

1. The expenditure of Q in 2022 is USD 115.2 mn.
Choice (A)

2. R incurred the highest expenditure in 2020.
Choice (C)

3. The second highest profit percentage in 2021 is $\frac{215-100}{100}$
$= 1.15$ or 115% Choice (B)

4. The profit of T in 2020 = USD 45 mn.
Three companies, P, Q and R, had a higher profit than T in 2020. Choice (D)

5. The company with the third highest profit percentage in 2022 is Q.
The profit of Q in 2022 = USD 74.8 mn.
Choice (A)

Solutions for questions 6 to 10:

At 9:00 AM, Kishan fills containers 1, 2, 4 and 9.
If there are no defective containers, at 11:00 AM, Ratan will fill containers 2, 3 and 5.
At 1:00 PM, Latif will fill containers 4, 5, 6, 7 and 8.
At 3:00 PM, Pavan will fill containers 1, 2, 4, 5, 7 and 8.
The following figure provides the persons who fill each container provided there are no defective containers.

K, P	K, R, P	R
K, L, P	R, L, P	L
L, P	L, P	K

6. There are three containers – 2, 4 and 5 – which have more than 40 litres of water (if there were no defective containers).
If we consider that only one container is defective, then because of this defective container at least one of the four persons should not fill containers 2, 4 and 5. Each container by being defective will only affect the water being filled in a maximum of three containers – the one to the right, the one below and the one to the left. However, there is no single container which will affect water being poured into all of 2, 4 and 5. Hence, it is not possible for only one container to be defective.
If there are two containers that are defective, there are multiple ways in which we can get the required result.
Containers 2 and 4 can be defective OR containers 4 and 5 can be defective and so on.
Hence, the minimum number of containers that must be defective is 2. Choice (B)

7. Given that container 8 was not filled by any person. From the above figure, we can see that Latif and Pavan would have filled container 8, if no container was defective. For Pavan to not fill container 8, container 9 must have been empty. Further, Kiran would definitely have filled container 9. Hence, container 9 must be defective.
Further, for Latif to not fill container 8, container 5 can be defective. In this case, Kiran will fill containers 1, 2, 4 and 9. Ratan will fill containers 2, 3 and 5. However, since container 5 is defective, this will become empty. Latif will fill containers 4, 5, 6, 7 and 9. Hence, if containers 5 and 9 are defective, no one will fill container 8.
However, even if containers 4 and 9 are defective, Ratan will fill containers 2 and 3. Latif will fill only containers 4, 5 and 6 (of which 4 is defective). Pavan will fill containers 1, 2, 4 and 5. Hence, even in this case, no one will fill container 8.
Hence, only container 9 will be definitely defective.
Choice (C)

8. At 2:00 PM, everyone except Pavan would have filled water in the containers.
If none of the containers were defective, the four persons would have filled water in the containers in the following manner:

K	K, R	R
K, L	R, L	L
L	L	K

In this case, each defective container affects a maximum of two containers – the container to the bottom and the container to the right.
There are 3 containers which have 40 litres of water – containers 2, 4 and 5.
For all these 3 containers to be affected, one of them must be defective. Even then it is not possible for any one container to be defective and get the required result.
Hence, at least two containers must be defective.
If containers 1 and 2 are defective, then all the non-defective containers will have 20 litres of water. Choice (A)

9. If containers 1, 2, 4 and 9 are defective, then none of the persons who fill the containers after Kiran will fill any container (because none of the containers will be filled when each of the next three persons try to fill the water).
In this case, 5 non-defective containers will be empty.
This cannot be more than 5 because Kiran will definitely fill water in 4 containers and if any of these are not defective, then other containers surrounding this non-defective container will be filled by any of the next three persons. Hence, the maximum number of non-defective containers that can be empty is 5. Choice (D)

10. From the figure above, we can see that if none of the containers are defective, then containers 3, 6 and 9 will have only 20 litres of water.

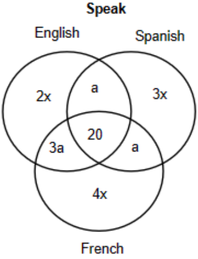
From the given condition, all the non-defective containers must have 40 litres of water. Hence, these three containers must definitely be defective. If these three containers are defective, the four persons will fill the containers as shown below:

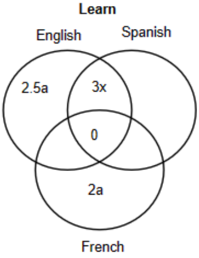
K, P		K, R		
K, L, P		R, L		
L, P		L		

We can see that if only containers 3, 6 and 9 are defective, container 8 will have only 20 litres of water. Hence, container 8 must also be defective. If container 8 is defective, Pavan will not fill container 7 (as container 8 is empty). Hence, container 7 must also be defective.
Now, container 4 has 60 litres of water. For this container to have only 40 litres of water, and for all the rest to have the same quantity of water, the only option is for container 5 to be defective. In this case, all the remaining containers will have 40 litres of water.
Hence, if containers 3, 5, 6, 7, 8 and 9 are defective, all the non-defective containers will have exactly 40 litres of water. Choice (D)

Solutions for questions 11 to 15:

Let the number of students who speak only English be 2x. From (i), the number of students who speak only Spanish is 3x and the number of students who speak only French is 4x.
From (vi), considering only the students who can speak exactly two languages, let a be the number of students who can speak only English and Spanish, b be the number of students who can speak only Spanish and French and c be the number of students who can speak only English and French.
In this group, number of students who can speak English = a + c
Number of students who can speak Spanish = a + b
Number of students who can speak French = b + c
From (vi), (a + c) = 2(a + b) and a + c = b + c ⇒ a = b.
Hence, c = 3a.
Number of students who can speak only English and Spanish = a
From (iii), number of students who want to learn only French = 2a
Number of students who can speak only one language = 2x + 3x + 4x = 9x
From (iv), number of students who want to learn English and Spanish = 3x
Also, for any student to want to learn all the three languages, he/she must not speak even one language. However, it is given that each student can speak at least one language. ∴ No student will want to learn all the three languages.
∴ Number of students who want to learn only English and Spanish = Number of students who want to learn English and Spanish = 3x
Number of students who can speak exactly two languages = a + a + 3a = 5a
From (v), number of students who want to learn only English = 2.5a
The first Venn diagram provides the number of students who can speak different combinations of the three languages, while the second Venn diagram provides the number of students who want to learn the three languages.





Also, the number of students who want to learn French must be less than the number of students who cannot speak French.
Since at least 2a students want to learn French and (2x + a + 3x) = (5x + a) students cannot speak French,
$2a < 5x + a \Rightarrow a < 5x$

Also, the number of students who want to learn English must be less than the number of students who cannot speak English.

Hence, $2.5a + 3x < 3x + a + 4x \Rightarrow 1.5a < 4x \Rightarrow a < \frac{8x}{3}$

The total number of students in the college = 250.

From the first Venn diagram, we get

$2x + a + 3x + 3a + 20 + a + 4x = 250 \Rightarrow 9x + 5a = 230$

The possibilities for (x, a) are (25, 1), (20, 10), (15, 19), (10, 28), (5, 37), (0, 46).

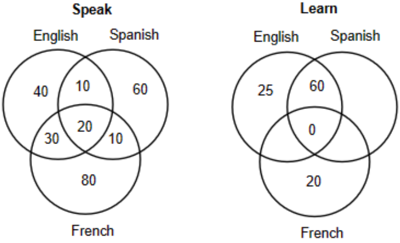
a cannot be odd because 2.5a number of students want to learn English and this cannot be a fraction.

Hence, we can ignore the possibilities where a is odd. Also, x cannot be 0 since the ratios given in the information will not hold.

The remaining possibilities are (20, 10), (10, 28). Of these two possibilities, in the second case, a is greater than 8x/3 (= 8× 10/3 = 26.67).

Hence, only one possibility exists for x and a, i.e., x = 20 and a = 10.

The following Venn diagram provides the values:



11. The number of students who speak only English is 40. Ans: (40)

12. The number of students who want to learn only French is 20. Ans: (20)

13. The students who cannot speak Spanish can learn to speak Spanish.
The number of students who cannot speak Spanish = 40 + 30 + 80 = 150.
Of these 150 students, 60 students want to learn both Spanish and English.
Hence, the maximum number of students who want to learn only Spanish is 150 – 60 = 90.
Ans: (90)

14. The number of students who can speak French = 140.
Of these 140 students, 50 students can speak English. Hence, the remaining 90 students can learn English.
However, the students who want to learn English and can speak French is at most 85, since any student who speaks French will not want to learn French.
A maximum of 85 students can speak French and want to learn English. Ans: (85)

15. The number of students who can speak either English or French = 190
All of these 190 students cannot learn both English and French, since they know at least one of the two languages.
Hence, only the remaining 60 students can learn both English and French. Choice (D)

Solutions for questions 16 to 20:

Let x be the number of patients who suffered from fever on a certain day.
From (i), the number of patients who suffered from stomach ache is 2x.
From (iii), the number of patients who suffered from headache is 2x + 10.
From (ii), the number of patients who suffered from cold is 2x + 15.
None of these can be the ailment mentioned in (v), because if one of these numbers is the same across the seven days, each of the other will also be same across the seven days.
Hence, the only ailment that (v) can refer to is cough.
Also, from (iv), the highest number of patients suffered from cough, for any day.
On any day, the number of patients who suffered from fever, stomach ache, headache and cold = 7x + 25
The minimum value of this is 32, for x = 1.
Given that on Day 1 and Day 2, 60 patients visited the clinic.
The maximum number of patients who suffered from cough on each of these two days = 60 – 32 = 28.

The number of patients who suffered from fever, stomach ache, headache and cold, when x = 2, is 7 × 2 + 25 = 39.
In this case, the number of patients who suffered from cough will be 21.
For Day 1 and Day 2, the value of x cannot be more than 2 because the number of patients who suffered from cough will not be the highest if x is more than 2.
For the other days, the number of patients who visited on each day = $\frac{560-120}{5} = 88$.
The number of persons who suffered from cough = 88 – 7x – 25 = 63 – 7x
This must be greater than the number of patients suffering from cold, i.e., 2x + 15.
$63 - 7x > 2x + 15 \Rightarrow x < 5.33$.
For x = 5, 7x + 25 = 60.
The number of patients suffering from cold = 2x + 15 = 25.
The number of patients suffering from cough = 28.
This is the minimum number of patients who suffered from cough on each day from Day 3 to Day 7.
Since this has to be the same across the seven days, the number of patients suffering from cough on each day must be 28.
The following table provides the number of patients suffering from each ailment, who visited the clinic on each day:

Day	Headache	Stomach Ache	Cough	Cold	Fever	Total
1	12	2	28	17	1	60
2	12	2	28	17	1	60
3	20	10	28	25	5	88
4	20	10	28	25	5	88
5	20	10	28	25	5	88
6	20	10	28	25	5	88
7	20	10	28	25	5	88

16. 25 patients suffering from cold visited the clinic on Day 4. Choice (A)

17. The total number of patients suffering from headache who visited the clinic = 124 Ans: (124)

18. The number of patients suffering from stomach ache who visited the clinic on Day 2 = 2
The number of patients suffering from headache who visited the clinic on Day 7 = 20
Required difference = 18. Ans: (18)

19. The sum of the total number of patients suffering from cold and the total number of patients suffering from fever = 159 + 27 = 186. Choice (A)

20. The difference between the total number of patients suffering from ailment X and the total number of patients suffering from ailment Y is maximum if X and Y are cough (196) and fever (27). For any other pair of ailments, the difference will be less than 196 – 27 = 169. Choice (D)

Difficulty level wise summary - Section II

Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	6, 7, 9, 16, 17, 18, 19, 20
Difficult	1, 2, 3, 4, 5, 8, 10, 11, 12, 13, 14, 15
Very Difficult	–

SECTION – III

Solutions for questions 1 to 22:

1. Let the first three terms be a, b, c respectively
∴ The fourth is a + c – b and the fifth is b + (a + c – b) – c = a
The sixth is third + fifth – fourth = c + a – (a + c – b) = b
The seventh is c and the eighth is a + c – b
Thus the given sequence is a periodic one, with a period of 4. The 3rd term is 4 and the 14th (∴ the 2nd) term is 7. Let the first term be a. The fourth has to be a – 3 and the sum of the first four is 2(a + 4)
The sum of the first 18 terms
= 4 (sum of the first four) + first + second
= 4[2(a + 4)] + (a + 7) = 57 given
∴ a = 2 and the fourth term is – 1.
The sum of the first 23 terms
= 5(sum of the first four) + (2 + 7 + 4)
= 5(2 + 7 + 4 –1) + (2 + 7 + 4)
= 5(12) + 13 = 73 Ans: (73)

2. As SR : QS = $1:\sqrt{2}$, PR : PQ = $1:\sqrt{2}$

As ∠QPR = 45° and $\frac{PR}{PQ}=\frac{1}{\sqrt{2}}=\cos 45°$.

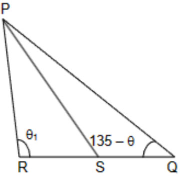
Alternately, if ∠PQR = θ, angle ∠PQR will be (135 – θ) and $\frac{\sin\angle PRQ}{\sin\angle PQR}=\frac{\sqrt{2}}{1}$

$\Rightarrow \frac{\sin\theta}{\sin(135-\theta)}=\frac{\sqrt{2}}{1}$

$\Rightarrow \frac{\sin\theta}{\sqrt{2}}=\sin 135°\cos\theta-\cos 135°\sin\theta$

$=\frac{\cos\theta}{\sqrt{2}}+\frac{\sin\theta}{\sqrt{2}} \Rightarrow \cos\theta = 0°$

⇒ ∠PRQ = 90°



⇒ ∠PRQ = 90°

That is $\frac{\left(\frac{40}{x}-\frac{30}{x}\right)}{\left(1+\frac{40}{x}\cdot\frac{30}{x}\right)}$ is maximum.

$\Rightarrow \frac{10}{x}\times\frac{x^2}{x^2+1200}$ is maximum.

$\Rightarrow \frac{10x}{x^2+1200}$ is maximum

$\Rightarrow \frac{x^2+1200}{10x}$ is minimum

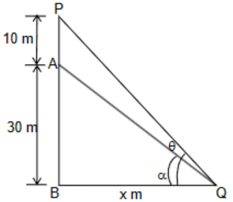
$\Rightarrow \frac{x}{10}+\frac{1200}{10x}$ is a minimum.

Now, since, product of $\left(\frac{x}{10}\right)$ and $\left(\frac{1200}{10x}\right)$ is constant, the minimum is achieved when $\frac{x}{10}=\frac{1200}{10x} \Rightarrow x^2 = 1200$ or $x=\sqrt{1200}=20\sqrt{3}$

$\therefore \angle PSR = 180^\circ - (90^\circ + 22\tfrac{1}{2}^\circ) = 67\tfrac{1}{2}^\circ$.
$\angle PQR = 180^\circ - (45^\circ + 90^\circ) = 45^\circ$
$\therefore \angle PSR + \angle PQR = 112\tfrac{1}{2}^\circ$. Choice (B)

3. $|x - a|$ represents the distance of x from 'a' on the number line.
$D = |x - 1| + |x - 2| + |x - 3|$ represents the sum of the distances of x from 1, 2 and 3. Clearly this is least when $x = 2$. i.e., $d = 2$. But as x moves away to either the left or the right from $x = 2$, d increases. By simple observation $d = 6$ for $x = 0$ and $x = 4$. Further for $x < 0$ and $x > 4$, $d > 6$.
Hence $0 \le x \le 4$. Choice (C)

4.



Let AB be the building and AP be the tower.
Let Q be a point on the ground at a distance of x m.
The angles of elevation of points P and A are θ and α respectively.
⇒ Angle subtended by tower = $\theta - \alpha$

$\tan\theta = \frac{10+30}{x} = \frac{40}{x}$ and $\tan\alpha = \frac{30}{x}$

Now $(\theta - \alpha)$ is maximum when $\tan(\theta - \alpha)$ is maximum.

For $\tan(\theta - \alpha)$ to be maximum, $\frac{\tan\theta - \tan\alpha}{1 + \tan\theta\tan\alpha}$ must be maximum.

$\therefore \sqrt{3}x = 60$ Ans: (60)

5. 20% of the cartons = 3 cartons
10% of pencil boxes in 3 cartons = 10% × 3 × 120 = 54
10% of the pencils in 54 pencil boxes = 81
Choice (A)

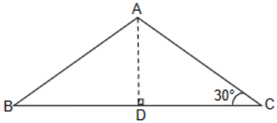
6. Let the length, breadth and the height of the cuboid be l, b and h.
$lbh = 64$
As the product of l, b and h is constant, their sum will be minimum when $l = b = h$.
When $l = b = h$, $l^3 = 64 \Rightarrow l = 4$
Hence, the sum of the edges of the cuboid $= 4(l + b + h) = 48$ units. Ans: (48)

7. Given that ABC = 5(AB + BC + CA)
$\Rightarrow 100A + 10B + C = 5[10A + B + 10B + C + 10C + A]$
$\Rightarrow 100A + 10B + C = 55A + 55B + 55C$
$\Rightarrow 45A = 45B + 54C \Rightarrow 5A = 5B + 6C$

$A = \frac{5B + 6C}{5} \Rightarrow A = B + \frac{6C}{5}$

So 'C' must be divisible by 5. Hence it can be 0 or 5
But $C \neq 0$, because CA is a two-digit number.
Hence, $C = 5$
∴ B can take 1, 2 and 3 and A will be 7, 8 and 9 respectively (i.e., 715, 825 and 935)
Hence, 3 such numbers exist. Ans: (3)

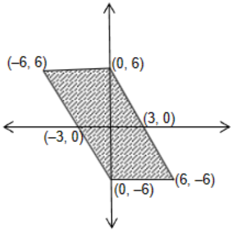
8. Let the value of the greatest angle be *x* degrees.
$\Rightarrow x = 2(180 - x) \Rightarrow 3x = 360 \quad \therefore x = 120$
The required ratio will be maximum when the other two sides are equal (because the smallest side needs to be as long as possible, i.e., while still remaining the shortest side)
⇒ the angle opposite the smallest side will be 30°.
Now, the triangle can be drawn as below:



Required ratio = $\frac{AC}{BC}$.

Consider $DC = \frac{BC}{2}$, and $\frac{DC}{AC} = \cos 30^\circ$ (in ΔADC).

$\Rightarrow \frac{DC}{AC} = \frac{\sqrt{3}}{2}$



$\Rightarrow \frac{AC}{BC} = \frac{1}{\sqrt{3}}$, i.e., $1 : \sqrt{3}$

Alternatively, once all the three angles are known to be 120°, 30°, 30°, we can use the sine rule to find the required ratio,

i.e., $\frac{AC}{BC} = \sin 30° : \sin 120°$

$= \frac{1}{2} : \frac{\sqrt{3}}{2} = 1 : \sqrt{3}$. Choice (B)

9. Since $|x| = x$ for $x \geq 0$

$-x$ for $x < 0$,

to find the region bounded by $|x| + |x + y| = 6$, we need to find the equivalent of $|x| + |x + y|$ = 6 in each quadrant.

In quadrant-1: $x > 0, y > 0$

$|x| + |x + y| = 6 \Rightarrow x + x + y = 6 \Rightarrow 2x + y = 6$

In quadrant-2: $x < 0, y > 0$

Case I: $-x > y$

$\Rightarrow |x| + |x + y| = 6$

$\Rightarrow -x + (x + y) = 6$

$2x + y = -6$

Case I : $x > y \Rightarrow |x| + |x + y| = 6$

$\Rightarrow -x + (x + y) = 6 \Rightarrow y = 6$.

Similarly, in quadrant-3: $x < 0, y < 0$

We get the equivalent of $|x| + |x + y|$ = 6 is

$2x + y = 6$

In quadrant-4: $x > 0, y > 0$

We get the equivalent of $|x| + |x + y|$ = 6 is

$y = -6$ for $-y > x$ and $2x + y = 6$ for $x > y$

By plotting these equations in the graph, the region bounded by them will look as follows.

It can be observed the bounded region is in the shape of parallelogram.

∴ The area of the bounded region = bh

= (6) (6 – (–6))

= 72 sq.units. Ans: (72)

10. Arranging the given observations in an increasing order, we get 4078, 4082, 4086, 4090, 4094, 4098, 4102, the observations are in an arithmetic progression.

$\therefore$ S.D (σ) = $|d|\sqrt{\frac{n^2-1}{12}}$; d is the common difference

$= 4 \cdot \sqrt{\frac{7^2-1}{12}} = 8$ Choice (B)

11. Let his initial speed be a m/s.

His speed for every subsequent 100 m will be $1.1a$, $(1.1)^2a$, $(1.1)^3a$ and $(1.1)^4a$.

Time taken to finish the race

$$= \frac{100}{a} + \frac{100}{1.1a} + \frac{100}{(1.1^2)a} + \ldots \frac{100}{(1.1^4)a}$$

$$= \frac{100}{a}\left(1 + \frac{1}{1.1} + \frac{1}{1.1^2} + \frac{1}{1.1^3} + \frac{1}{1.1^4}\right)$$

$$= \frac{100}{a} \times \frac{\left[1\left(1 - \left(\frac{1}{1.1}\right)^5\right)\right]}{\left(1 - \frac{1}{1.1}\right)}$$ (using sum to n terms in GP)

$$= \frac{100}{a} \times \left(\frac{0.379}{0.091}\right) = \frac{100}{a}(4.165)$$

Given, the time is 194 seconds.

$\frac{100}{a}(4.165) = 194$

$\Rightarrow a$ = 2.147 m/s, i.e., 2.147 × $\frac{18}{5}$ = 7.73 kmph.

Choice (A)

12. The minimum value will be realised when $x = y = z = {}^1/_3$ and will be $(1/3 + 3)^2$ (3) $= (10/3)^2$ (3) = 100/3 = $33^1/_3$.

If x, y, z are not taken as equal, say x = 0.3, y = 0.3, z = 0.4 the value of the expression is

$2\left(\frac{3}{10}+\frac{10}{3}\right)^2+\left(\frac{2}{5}+\frac{5}{2}\right)^2 \approx 34.5$

∴ The minimum value is $33^1/_3$. Choice (C)

13. We can assume that idly, puri dosa, upma and vada correspond to A, B, C, D, E respectively. The question asks for the number of derangements of these 5 objects. The number of ways of assigning these 5 objects to the 5 people, so that each of them goes to the 'wrong' person (or each person gets a 'wrong' item) is the number of derangements of 5 objects.

This is $5!\left(\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}\right)$

= 3(4)(5) – 4(5) + 5 – 1 = 44.

Ans: (44)

14. Let the base of the number system be *n* and the two-digit number be *ab*.
Given $2(ab)_n = (ba)_n$
$\Rightarrow nb + a = 2na + 2b$
$\Rightarrow a(2n-1) = b(n-2)$
$\Rightarrow \frac{a}{b} = \frac{n-2}{2n-1}$

Now the maximum possible value of *a* or *b* is (*n* – 1) (i.e., in base *n*)

for *n* = 3 we get $\frac{a}{b} - \frac{1}{5}$, which is not possible.

Similarly, *n* = 4 is not possible, but for *n* = 5

We get $\frac{a}{b} - \frac{3}{9}$, i.e., $\frac{1}{3}$. Hence, the least possible value of *n* is 5.

∴ *a* = 1, *b* = 3 is the only number possible when *n* is the least possible.
∴ The correct wage = $(13)_5 = (8)_{10}$
The actual wage paid = $(31)_5 = (16)_{10}$
∴ Excess wage paid = $(8)_{10}$ Ans: (8)

15. $\left(7+4\sqrt{3}\right)^{20} = 7^{20} + 20_{C_1}7^{19}(4\sqrt{3})^1 + 20_{C_2}7^{18}(4\sqrt{3})^2 + \ldots + 20_{C_{19}}(7)(4\sqrt{3})^{19} + (4\sqrt{3})^{20}$

$\left(7-4\sqrt{3}\right)^{20} = 7^{20} - 20_{C_1}7^{19}(4\sqrt{3})^1 + 20_{C_2}7^{18}(4\sqrt{3})^2 + \ldots - 20_{C_{19}}(7)(4\sqrt{3})^{19} + (4\sqrt{3})^{20}$

$\therefore \left(7+4\sqrt{3}\right)^{20} + \left(7-4\sqrt{3}\right)^{20} - 2\left[7^{20} + 20_{C_2}(7)^{18}(4\sqrt{3})^2 + \ldots + (4\sqrt{3})^{20}\right]$

as $\left(\sqrt{3}\right)^{\text{Evenno.}} = \text{Integer}$

$\therefore \left(7+4\sqrt{3}\right)^{20} + \left(7-4\sqrt{3}\right)^{20} -$ I + F + *f* = an integer
∵ I is an integer, (F + *f*) must also be an integer
  0 < F < 1
  0 < *f* < 1
  0 < F + *f* < 2
∴ F + *f* must be 1 and 42(F + *f*) = 42. Ans: (42)

16. Since A and B worked for the same amount of time and A is twice as efficient as B, A's share will be twice that of B. Hence, C's share = 18000 – (5100 + 10200) = 2700.

Hence, C did $\left(\frac{2700}{18000}\right)^{\text{th}}$ part of the job.

$\Rightarrow \frac{9}{60}$ th of the work is completed by C on the 1st day.

$\therefore$ % efficiency of C = $\frac{\left(\frac{9}{60}\right)}{\left(\frac{1}{5}\right)} \times 100 = 75\%$.

Choice (C)

17. $\log_5 \log_4 \log_3 \log_2 \left(a^{b^c}\right) = 0$

$\Rightarrow \log_3 \log_2 \left(a^{b^c}\right) = 4 \Rightarrow a^{b^c} = 2^{3^4}$

If we consider corresponding values of a, b, c then a + b + c = 2 + 3 + 4 = 9

But we can write the following as $a^{b^c} = 2^{3^4}$

$= 2^{9^2} = 8^{3^3}$
Correspondingly, the maximum value of the sum will be a + b + c = 8 + 3 + 3 = 14. Choice (D)

18. Given N = (35)(36) …….(67).

We need the value of $\text{Rem}\left[\frac{N}{289}\right]$

$\therefore \text{Rem}\left[\frac{N}{289}\right] = 17\ \text{Rem}\left[\frac{(1)(2)\ldots.(16)(3)(1)(2)\ldots.(16)}{17}\right]$

[∵ 51= 17(3)]
= 17(–1) 3(–1) = 51. Choice (C)

19. There are 3 steps along the North-South direction and 6 along the East-West direction. Hence, number of ways required = $\frac{(6+3)!}{3!\,6!} - 84$.

Choice (B)

20. The index of the highest power of 5 in N! is 64.

$\left[\frac{N}{5}\right]+\left[\frac{N}{5^2}\right]+\left[\frac{N}{5^3}\right]=64$ [We do not consider $\left[\frac{N}{5^4}\right]$, for the number N is definitely less than 625.]

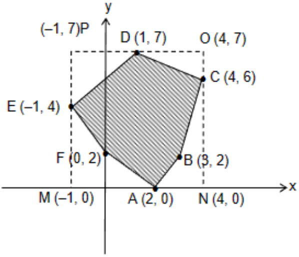
If $\frac{N}{5^3}$ = K, then N = 125K

$\frac{N}{5}+\frac{N}{5^2}+\frac{N}{5^3}$ = 25K + 5K + K = 31K. For K = 2, we get N = 250. The highest power of 5 that divides 250! is 5^{62}.
So, N! has 64 fives means the minimum value of N is 260 where as maximum value of N is 264.
Similarly for M = 125, the number of 5's in 125! is 31.
We see that the number 125 = 5^3. So 124! will have 31 – 3 or 28 fives. The minimum value of M is 120.
So, the maximum difference between the values of M and N is 264 – 120 = 144.

Alternative solution:

To arrive N, such that the highest power of 5 in N!, is say k, start with N as approximately four times k.
Hence, if the highest power of 5 in N! is 64, then check for N = 4 × 64 = 256.
Now, the highest power of 5 in 256! is 63, hence, we need to move the next multiple of 5 (i.e., 260) to get the highest power as 64. And this highest power will remain as 64, for N = 260, 261, 262, 263 and also 264. Thus the maximum possible value of N = 264.
Similarly, we can see that the minimum possible value of M, such that highest power of 5 in M! is 28, is 120.
Thus the maximum difference between M and N is 264 – 120 = 144. Ans: (144)

21. The given points can be represented on the x-y plane as follows:



The given polygon is ABCDEF
Let MNOP be a rectangle that circumscribes the given polygon.
Required Area = Area of Rectangle MNOP – Area of (ΔPED + ΔEFM + ΔMFA + ΔABN + ΔNBC + ΔCDO)

$= PO \times ON - \frac{1}{2}$ [(PD)(PE) + (EF)(Altitude from F to EF) + (MA)(Altitude from F to MA) + (AN)(Altitude from B to AN) + (NC) (Altitude from B to NC) + (DO) (OC)]

$= 5 \times 7 - \frac{1}{2}[(2)(3) + (4)(1) + (3)(2) + (2)(2) + (6)(1) + (3)(1)]$

$= 35 - \frac{1}{2}(29) = 35 - 14.5$

= 20.5 sq.units.

Alternative Solution:
Area of the polygon with vertices as (x_1, y_1), (x_2, y_2), ... (x_6, y_6)

$$= \frac{1}{2}\begin{vmatrix} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_1 \\ y_1 & y_2 & y_3 & y_4 & y_5 & y_6 & y_1 \end{vmatrix}$$

$$= \frac{1}{2}\left[\begin{matrix}(x_1y_2 - y_1x_2) + (x_2y_3 - y_2x_3) + (x_3y_4 - y_3x_4) + \\ (x_4y_5 - y_4x_5) + (x_5y_6 - y_5x_6) + (x_6y_1 - y_1x_6))\end{matrix}\right]$$

$$= \frac{1}{2}\begin{vmatrix} 2 & 3 & 4 & 1 & -1 & 0 & 2 \\ 0 & 2 & 6 & 7 & 4 & 2 & 0 \end{vmatrix}$$

$$= \frac{1}{2}\left|(4-0)+(18-8)+(28-6)+(4+7)+(-2)+(-4)\right|$$

= 20.5 sq.units Ans: (20.5)

22. Given, $-y = \dfrac{x^2 + 3\sqrt{x}\,(2x+9) + 162}{x + 9\sqrt{x} + 18}$

$= \dfrac{(x^{3/2}+27)(\sqrt{x}+6)}{(\sqrt{x}+3)(\sqrt{x}+6)} = x - 3\sqrt{x} + 9$ for $16 \le x \le 81$

$\Rightarrow 13 \le -y \le 63;$

$\Rightarrow -63 \le y \le -13$ Choice (D)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	–
Easy	5, 10
Medium	3, 6, 7, 12, 16, 17, 19, 20, 21, 22
Difficult	1, 2, 8, 11, 13, 14
Very Difficult	4, 9, 15, 18

Done
Tip: Scroll slowly for the best results